nafeabubaker@gmail.com
+970 597785625

Nafe Abubaker

GitHub
LinkedIn
Portfolio

Overview

Computer Engineering graduate with practical experience in backend development, data engineering, and applied AI. Built REST APIs with C#/.NET and FastAPI, automated ETL workflows, and developed NLP, computer vision, and RAG applications in Python. Currently an E-Channels Officer at Bank of Palestine and an AI Engineer Intern at Cellula Technologies, combining technical experience with an understanding of customer needs and business workflows. Earned the top final evaluation in the IBT x GGateway Data Science and Machine Learning bootcamp.

Professional Experience

AI Engineer Intern **Sep 2026 – Present**
Cellula Technologies ***Remote***

- Ongoing AI engineering internship focused on NLP, MLOps, computer vision, large language models (LLMs), and AI agents.
- Built and evaluated RNN and LSTM models for nine-class NLP classification using TensorFlow/Keras, combining user queries with image captions.
- Developed a RAG pipeline using PDF chunking, sentence embeddings, FAISS retrieval, and LLM generation; deployed NLP and document Q&A apps with Streamlit.

E-Channels Officer **May 2026 – Present**
Bank of Palestine ***On-site (Ramallah)***

- Identify recurring issues in customer and corporate banking workflows and communicate business needs to technical teams to improve usability.
- Perform user acceptance testing (UAT) for digital banking platforms, validating permissions and approval workflows and documenting functional and usability issues.

Business Support System Engineer **Jan 2026 – Apr 2026**
Jawwal ***On-site***

- Supported telecom business system operations and coordinated issue resolution with cross-functional teams to maintain reliable internal workflows.

Data Science & Machine Learning Trainee **Jan 2026 − Apr 2026**
IBT x GGateway **Remote**

- Progressed through an intensive end-to-end Data Science & ML bootcamp, from Python, SQL, data cleaning, EDA, and statistics to machine learning, deep learning, NLP, LLMs, and deployment.
- Applied the full workflow in a team capstone on student dropout prediction, contributing to EDA, data preparation, modeling, evaluation, and presentation; earned the bootcamp's top final evaluation.

Backend Engineering Trainee **Oct 2024 − Jan 2025**
ASAL Technologies **On-site (Rawabi)**

- Developed RESTful APIs in C#/.NET Core with MongoDB CRUD operations.
- Implemented authentication, validation, and collaborated with cross-functional teams on backend integration.

Data Engineering Trainee **Feb 2025 − Jun 2025**
ASAL Technologies **On-site (Rawabi)**

- Built an event-driven Prefect 2.0 ETL pipeline triggered by MongoDB receipt inserts, calculating operational KPIs and idempotently upserting results into four MySQL analytics tables.
- Exposed four analytics datasets through asynchronous FastAPI endpoints using SQLAlchemy and Pydantic, covering receipt summaries, monthly store sales, item performance, and peak hours.

Education

Birzeit University B.Sc. in Computer Engineering – GPA: 3.2*/4.0* **Ramallah** **2020 − 2025**

Projects

Student Dropout Prediction [GitHub] **Apr 2026**

- Coordinated the project and led data and EDA work for a recall-prioritized dropout-risk model using the HSLS:09 dataset of approximately 15,900 students.
- Compared Logistic Regression, Random Forest, XGBoost, and LightGBM; a tuned XGBoost model at a 0.30 threshold achieved 86.5% recall, identifying 423 of 489 dropouts in the test set
- **Technologies**: Python, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, Matplotlib, Seaborn.

Traffic Management & Red-Light Violation [GitHub] **Jul 2025**

- Built an end-to-end computer vision system for queue-aware traffic signal recommendations using overhead drone footage (YOLOv8 + SORT + polygon ROIs).
- Achieved 91.7% mAP@0.5 on the validation set for YOLOv8 vehicle detection, and improved traffic throughput by ~0.25 cars/s compared to fixed timing (simulation-based).
- Implemented red-light violation detection using stop-line logic and best-frame evidence capture, then extracted license plates using plate detection + OCR.
- **Technologies**: YOLOv8, SORT, OpenCV, OCR, Roboflow, FastAPI, React, MongoDB.

Palestine Store Hub [GitHub] **Jan 2025**

- Developed 21 REST endpoints across three ASP.NET Core controllers for users, stores, and products, supporting CRUD, search, filtering, geolocation-based discovery, ratings, and recommendations
- Integrated three MongoDB collections through dedicated service providers, implementing duplicate checks, partial updates, image validation, Swagger documentation, and CORS-based frontend integration.
- **Technologies**: C#/.NET Core, MongoDB, REST APIs.

Arabic Question Answering [GitHub] **Jan 2024**

- Built a simple Arabic Q&A demo using a pre-trained AraELECTRA question-answering model.
- Applied AraBERT-style preprocessing (ArabertPreprocessor) and tokenization to format question/context inputs and extract answer spans.
- **Technologies**: PyTorch, Transformers, AraBERT, NLP.